Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *MILU* | ***Context:*** *Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)*

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

This is a fundamental mismatch. MILU's output ontology is a single MCQ label — one of up to four options [Q33, Q55] — and the dataset schema confirms only a 'target' field with no rationale or explanation field [DATASET-D1, D9]. The deployment requires a correct/incorrect verdict PLUS a substantive Hindi-language explanatory rationale [elicitation Q3]. MILU has no rubric, no label set, and no scoring function for rationale quality. The 41-subject taxonomy [Q42] supports per-subject accuracy reporting but cannot validate explanation quality. Compounding this, the benchmark itself reports models perform worst in Arts & Humanities, Social Sciences, and Law & Governance [Q6, Q20, Q62, Q71, Q77] — precisely the deployment's priority subjects (History, Polity, GK).

ID	Evidence
Q33	"During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency." (p.4)
Q55	""For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., argmax(logP(a1\" (p.5)
Q3	"We evaluate over 42 LLMs, and find that current LLMs struggle with MILU, with GPT-4o achieving the highest average accuracy at 74%." (p.1)
Q42	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
Q6	"Domain-wise analysis indicates that models perform poorly in culturally relevant areas like Arts & Humanities and Law & Governance compared to general fields like STEM." (p.1)
Q20	"Our domain-wise analysis reveals that models perform poorly in culturally relevant areas, such as Arts & Humanities and Social Sciences, compared to more general fields like STEM." (p.2)
Q62	"When analyzed across domains, the models generally perform worse in Arts, Humanities, and Social Sciences than in STEM subjects." (p.6)
Q71	"We analyze the performance of various base and instruct models across multiple domains and languages. Similar trends to those in Section (§5.2) are observed where the open models perform poorly in domains specific to Indian culture—such as Arts & Humanities, Social Sciences, and Law & Governance—but demonstrate higher performance in STEM fields." (p.7)
Q77	"Additionally, the domain-specific analysis indicates that models perform better in general fields such as STEM while facing challenges in culturally relevant subjects like Arts, Humanities, and Law, highlighting the lack of this knowledge in the current models and datasets." (p.8)
D1	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया? … 44वें संविधान संशोधन अधिनियम द्वारा — Polity/Constitution question in Hindi, about 44th Constitutional Amendment — UPSC core topic
D9	If Rs. 4000 becomes Rs. 5760 in 2 years at compound interest … what is the annual rate of interest? … 20 percent — Mathematics/Quantitative Aptitude question (compound interest) — confirms presence in English partition

Strengths

- 41-subject × 8-domain hierarchical taxonomy [Q42] supports per-subject diagnostic accuracy reporting, useful for identifying weak areas.
- Closed-form MCQ accuracy is unambiguous for the verdict portion of the deployment's required output.

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Q42	“Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies.” (p.4)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: MCQ option labels are regionally relevant only insofar as the questions are India-sourced [Q26]. The output category space is, however, structurally incomplete for this deployment.

OO-2: Critical missing categories: rationale-quality dimensions (factual accuracy, fluency, pedagogical clarity, register appropriateness — listed in elicitation pedagogical_context). MILU has none of these.

OO-3: MCQ-only ontology does not encode non-regional values per se, but encodes an evaluation philosophy (option-selection only) that is misaligned with pedagogical-feedback deployments.

OO-4: Significant taxonomy redesign is required: a separate rationale-quality rubric must be developed, as MILU provides no foundation. The 2025 Hindi LLM benchmarking preprint [WEB-14] confirms no such Hindi rubric exists publicly.

OO-5: MILU's MCQ-only output ontology violates structural validity for this deployment (the construct of pedagogical feedback has internal structure — verdict + rationale — that MILU cannot represent) and content validity (rationale-quality categories are entirely absent).

ID	Evidence
Q55	“"For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., argmax(logP(a1\” (p.5)
Q26	“These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers.” (p.3)
Q33	“During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency.” (p.4)
WEB-14	arXiv:2508.19831 confirms Hindi instruction-following / conversational benchmarks 'largely unavailable publicly' (https://arxiv.org/html/2508.19831v1)
WEB-13	IndicGenBench (ACL 2024) covers Hindi generation but not exam rationale quality (https://aclanthology.org/2024.acl-long.595/)
Q42	“Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies.” (p.4)
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Q71	“We analyze the performance of various base and instruct models across multiple domains and languages. Similar trends to those in Section (§5.2) are observed where the open models perform poorly in domains specific to Indian culture—such as Arts & Humanities, Social Sciences, and Law & Governance—but demonstrate higher performance in STEM fields.” (p.7)

Requires Expert Verification

- A pedagogical-NLP expert should design a Hindi exam rationale rubric covering factual accuracy, fluency, and pedagogical clarity — this is wholly outside MILU's scope.

Remediation

Gap 1: No taxonomy or label set for rationale quality dimensions exists in MILU.

Recommendation: Adopt or develop a multi-axis rationale-quality taxonomy (e.g., 4-axis: factuality, fluency, pedagogical helpfulness, Manak Hindi register), informed by Hindi-medium coaching pedagogy conventions. Expert verification by Hindi-medium UPSC/SSC educators is required.